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May University in Cairo

Faculty of Engineering
Robotics, Autonomous and Interactive Systems
Measurement Systems and Sensors Course
2nd Level – 2nd Semester 2025-2026

Date: 2-3-2026

Project Proposal:

Smart Diabetic Foot Ulcer Detection and Prevention Shoe

Design and implement a smart wearable shoe insole that continuously monitors foot conditions using multiple sensors and actively prevents ulcer formation by redistributing pressure and providing corrective feedback.

Team Members:

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Submitted to:

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Background:

Diabetic foot ulcer is one of the most serious complications affecting diabetic patients and is a major cause of hospitalization and lower-limb amputation worldwide. Foot ulcers typically develop due to prolonged high plantar pressure, poor blood circulation, excessive moisture, and unnoticed inflammation caused by neuropathy.Patients often cannot feel pain or pressure abnormalities, which allows tissue damage to progress without early detection. Continuous clinical monitoring is not always practical outside hospitals, creating a need for wearable systems capable of measuring physiological and biomechanical parameters in real time.

Objectives:

- Design and build a smart insole capable of monitoring multiple ulcer risk factors.
- Measure plantar pressure distribution across the foot in real time.
- Detect localized temperature increase indicating early inflammation.
- Monitor humidity levels inside the shoe environment.
- Analyze gait and walking patterns using motion sensing.
- Implement a prevention system that reduces harmful pressure conditions.
- Estimate local blood perfusion using optical sensing.
- Develop a control algorithm that calculates a Foot Ulcer Risk Index.
- Provide alerts and feedback to the user when dangerous conditions occur.
- Ensure low-cost implementation using locally available components.

Scope:

The project will deliver a functional prototype of a Smart Foot Ulcer Detection and Prevention System including:

- Research on diabetic foot pathology and prevention techniques.
- Design of a wearable smart insole structure.
- Integration of pressure, temperature, humidity, motion, and perfusion sensors.
- Development of an active prevention mechanism using vibration feedback.
- Implementation of embedded control logic using a microcontroller.
- Wireless data transmission to a monitoring interface.
- Testing system performance under simulated walking conditions.
- Evaluation of measurement accuracy and response efficiency.
- The system focuses on prevention and early detection rather than medical diagnosis.

Timeframe:

	Description of Work	Start and End Dates
~~**Phase One**~~	~~Research, medical study, and system design~~	~~Week 3-4~~
~~**Phase Two**~~	~~Hardware prototyping and sensor integration~~	~~Week 6-7~~
Phase Three	Software development and control algorithms	Week 8-9
Phase Four	Testing, calibration, and final adjustments	Week 10

Project Components:

Project Components	Anticipated Cost
esp32	375- 400
Velostat sheet	90-100
Aluminum foil	30-60
NTC thermistor x4	12-20
DHT22 Humidity	275-290
MPU6050 IMU	175-185
MAX30102 Pulse Sensor	240-260
Coin Vibration Motor x3	60-90
Thermoelectric cooler	150-170
MOSFET IRLZ44N	55-60
Charging module TP4046	35-40
li-ion batteries x2	100-120
Battery housing	60-70
Control housing	60-70
Wires, Resistors, etc.	100-150
Shoe	450-600
Total	**1600-2700**

1. ESP32 Microcontroller

The **ESP32** is used as the main controller of the system.

Why chosen:

- Provides sufficient processing power to handle multiple sensors simultaneously.
- Built-in **Bluetooth and Wi-Fi** enable wireless monitoring and alerts without extra modules.
- Low power consumption, suitable for wearable medical devices.
- Widely available and inexpensive in the Egyptian electronics market.

Role in system:

Collects sensor data, processes risk conditions, and activates prevention mechanisms.

2. Velostat Sheet with Aluminum Foil (Pressure Sensor)

A **Velostat sheet** combined with aluminum foil layers is used as a flexible pressure sensor inside the shoe.

Why chosen:

- Much cheaper alternative to commercial FSR sensors.
- Flexible and comfortable for wearable footwear applications.
- Resistance changes with applied pressure, allowing detection of high plantar pressure.
- Easy to cut and customize for different foot regions.

Role in system:

Detects abnormal pressure points that may lead to foot ulcers.

3. NTC Thermistors (×4)

NTC thermistors are used to measure temperature at multiple foot locations.

Why chosen:

- High sensitivity to small temperature changes.
- Low cost compared to digital temperature arrays.
- Small size suitable for embedding inside footwear.
- Reliable for detecting localized inflammation.

Role in system:

Identifies early inflammation through temperature increase at pressure zones.

4. DHT22 Humidity Sensor

The **DHT22** measures humidity levels inside the shoe.

Why chosen:

- Provides accurate humidity readings compared to cheaper alternatives.
- Simple digital interface with ESP32.
- Detects moisture buildup, which increases infection risk.

Role in system:

Monitors foot environment conditions that promote bacterial growth.

5. MPU6050 IMU (Accelerometer + Gyroscope)

The MPU6050 measures motion and foot orientation.

Why chosen:

- Combines accelerometer and gyroscope in one compact module.
- Low cost and widely available.
- Enables gait and walking pattern analysis.

Role in system:

Detects abnormal walking patterns that create uneven pressure distribution.

6. MAX30102 Pulse Sensor

The MAX30102 measures heart rate and blood oxygen signals.

Why chosen:

- Optical sensor capable of detecting pulse variations related to blood circulation.
- Compact and low power.
- Commonly used in wearable health monitoring devices.

Role in system:

Helps estimate blood flow quality, which is critical in ulcer prevention.

7. Coin Vibration Motors (×3)

Small vibration motors provide haptic feedback to the user.

Why chosen:

- Lightweight and comfortable for wearable systems.
- Low power consumption.
- Immediate tactile alert without needing visual or audio signals.

Role in system:

Warns the user when dangerous pressure or temperature conditions are detected.

8. Thermoelectric Cooler (Peltier Module)

A thermoelectric cooler is used as an active prevention component.

Why chosen:

- Can reduce localized temperature electronically.
- Compact solid-state device with no moving parts.
- Helps prevent inflammation escalation.

Role in system:

Cools high-temperature areas to reduce tissue stress and ulcer formation risk.

G. IRLZ44N MOSFET

The IRLZ44N is used as a power switching device.

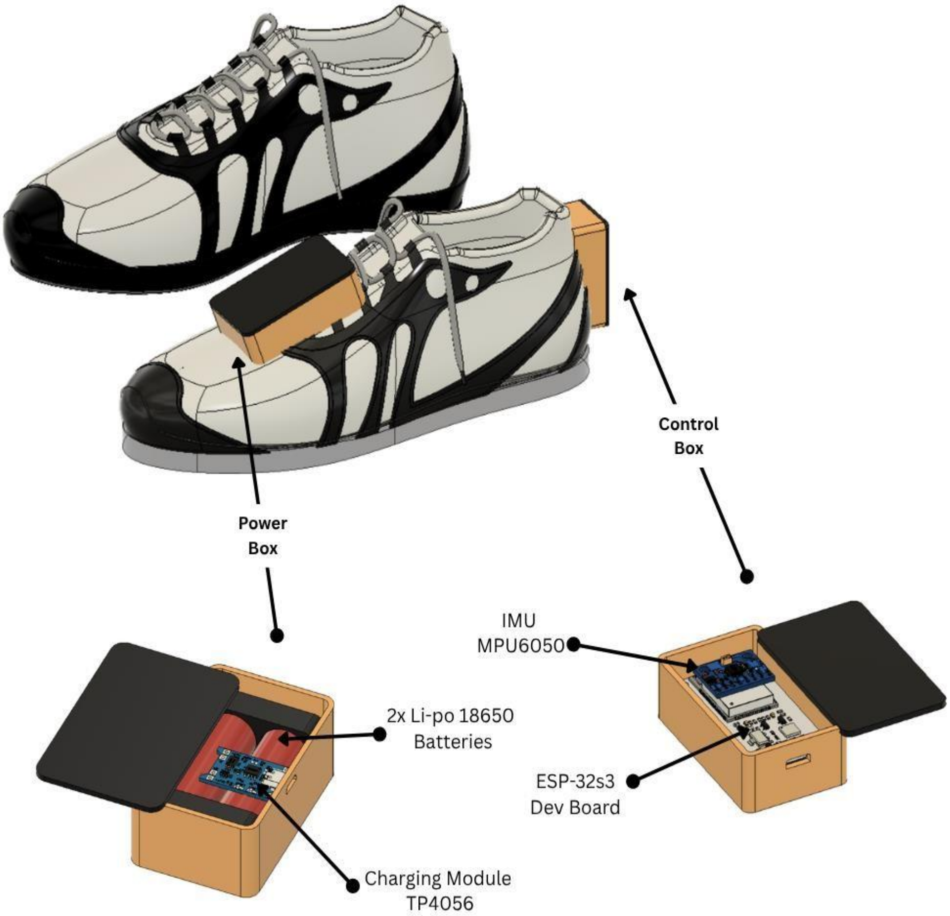
Why chosen:

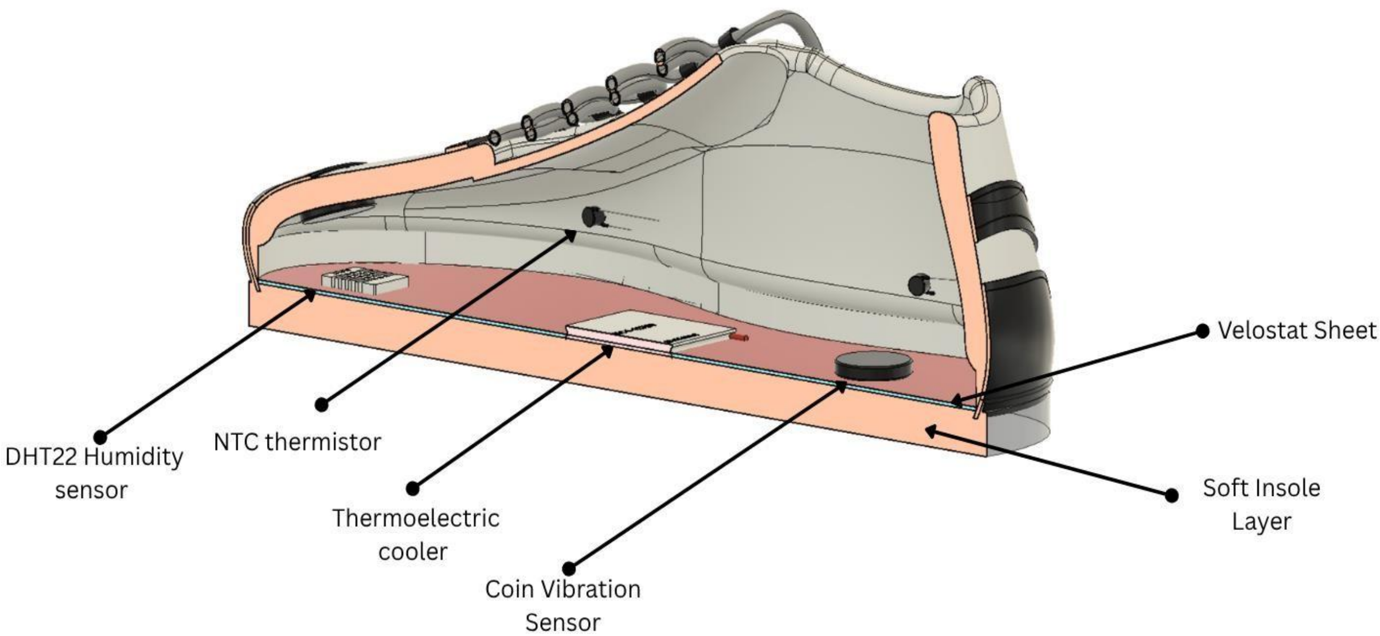
- Logic-level MOSFET compatible with ESP32 output voltage.
- Handles higher current required by vibration motors and thermoelectric cooler.
- Efficient switching with low heat loss.

Role in system:

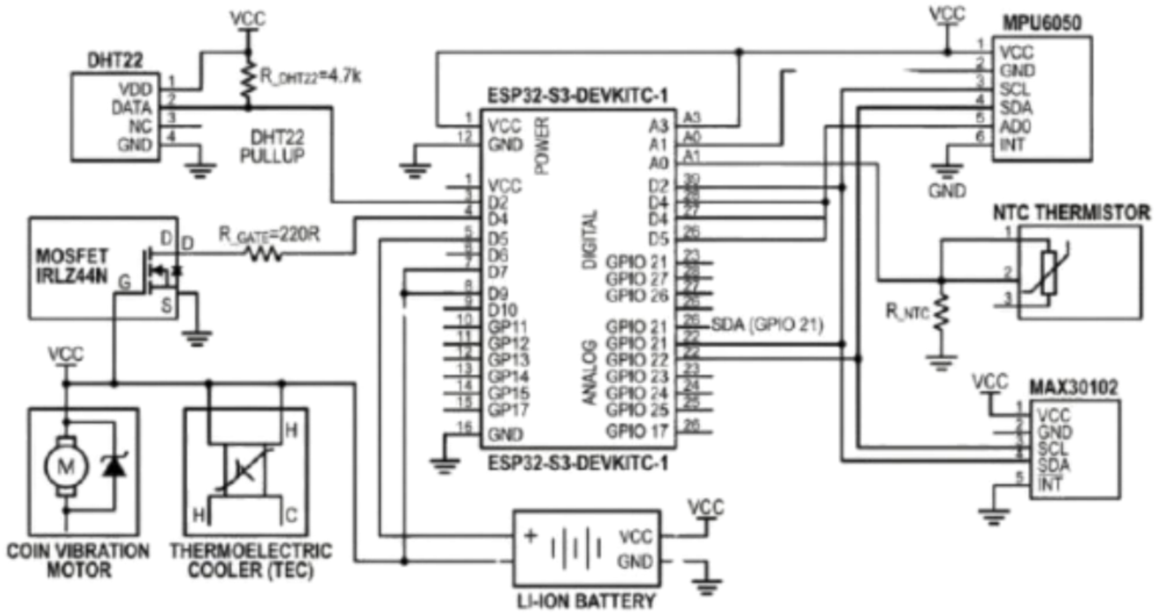
Allows the ESP32 to safely control high-power components.

Mechanical Design:





Circuit Diagram:



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- ScienceDirect – Plantar Pressure Monitoring Systems. (2024)
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